Stanley Supplementary Exercise 27(a)

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Fixed statement June 4, 2023 Fixed solution March 24, 2023

Original statement

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27.

(a) [3–] Define a linear transformation $\phi : \Lambda^n_{\mathbb{Q}} \to \Lambda^n_{\mathbb{Q}}$ by $\phi(e_\lambda) = m_{\lambda'}$. Find the size of the largest block in the Jordan canonical form of ϕ.

Original solution

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27.

(a) The matrix of ϕ^{-1} with respect to the basis $\{m_\lambda\}$ is the matrix M of Section 7.4. By a simple result of linear algebra, M and M^{-1} have the same Jordan block sizes. By Theorem 7.4.4 the matrix M is upper triangular with 1's on the main diagonal. Hence the size of the largest Jordan block of ϕ is the least integer m for which $(M-I)^m = 0$. Again by Theorem 7.4.4 we have $M_{\lambda\mu} = 0$ unless $\mu \leq \lambda'$. By Proposition 7.4.1 and the Gale-Ryser theorem (a basic combinatorial result which everyone should know) the converse is true, i.e., $M_{\lambda\mu} > 0$ if $\mu \leq \lambda'$. Since all entries of M are nonnegative, there is no cancellation in the computation of powers of $M - I$. It follows that m is the size of the longest chain in the poset $\mathrm{Par}(n)$, ordered by dominance.

Now use Exercise 7.2(f).

(This exercise gives the *length* of the longest chain, so you need to add 1 to get the size.) For some further work on chains in $\mathrm{Par}(n)$, see E. Early, *Discrete Math.* **313** (2013), 2168–2177.